CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Using Euclid's subtraction algorithm for gcd(360, 156), the number of subtraction steps needed is: 1

(a) 9 **(b)** 11 **(c)** 8 **(d)** 10

2. Using Euclid's subtraction algorithm for gcd(420, 364), the number of subtraction steps needed is: 1

(a) 10 **(b)** 11 **(c)** 8 **(d)** 9

3. A counter starts at 0. For i = 1, 2, …, 50, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

(a) 23 **(b)** 25 **(c)** 22 **(d)** 24

4. Set s = 0. While n > 0: add (n mod 10) to s, then replace n by the quotient of n divided by 10. Starting with n = 32481, the final s is: 1

(a) 19 **(b)** 20 **(c)** 17 **(d)** 18

5. Using gcd(20, 25) = 5, the LCM of 20 and 25 is: 1

(a) 95 **(b)** 105 **(c)** 100 **(d)** 120

6. To test whether 373 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{373}\rfloor$. The last number checked is: 1

(a) 18 **(b)** 186 **(c)** 20 **(d)** 19

7. Euclid's subtraction algorithm is applied to find gcd(312, 200). The number of subtractions performed (until a pair with 0 appears) is: 1

(a) 9 **(b)** 10 **(c)** 11 **(d)** 8

8. By Euclid's algorithm, gcd(453, 384) is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

9. Using Euclid's subtraction algorithm for gcd(294, 245), the number of subtraction steps needed is: 1

(a) 5 **(b)** 8 **(c)** 7 **(d)** 6

10. Using Euclid's subtraction algorithm for gcd(448, 176), the number of subtraction steps needed is: 1

(a) 8 **(b)** 11 **(c)** 10 **(d)** 9

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the gcd of 106 and 44 by Euclid's subtraction algorithm and use it to find their LCM. 2

12. To test whether 134 is prime, check divisibility by 2, 3, … up to $\lfloor\sqrt{134}\rfloor$. Carry out the test and give the conclusion. 2

13. Apply Euclid's subtraction algorithm to find gcd(224, 154). Write the sequence of pairs and count the subtractions. 2

14. Find the gcd of 111 and 74 by Euclid's subtraction algorithm and use it to find their LCM. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find gcd(84, 68) in two ways: 3
(i) by listing the divisors of both numbers,
(ii) by Euclid's subtraction algorithm. Check that the answers agree.

16. A counter starts at 0 and for i = 1, 2, …, 26 increases by 1 whenever i is a multiple of 3 or of 5. Find the final count without listing everything. 3

17. Trace: x = 23; repeat 5 times: x = 2x − 4. Write the value of x after each repetition and the final value. 3

OR

Find gcd(104, 65) in two ways: 3
(i) by listing the divisors of both numbers,
(ii) by Euclid's subtraction algorithm. Check that the answers agree.

18. A counter starts at 0 and for i = 1, 2, …, 34 increases by 1 whenever i is a multiple of 3 or of 5. Find the final count without listing everything. 3

OR

Trace: x = 20; repeat 3 times: x = 2x − 3. Write the value of x after each repetition and the final value. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Trace: r = 0; while n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Start with n = 1432. List r after each pass and say what the procedure does. 5

OR

Start with n = 9. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 5

20. Trace: r = 0; while n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Start with n = 1217. List r after each pass and say what the procedure does. 5

OR

Start with n = 6. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND